

PES-VCS: Version Control System

Operating Systems – Orange Problem (Unit – 4)

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Sem: 4 Sec: A

System Architecture

This project implements a simplified version control system inspired by Git. It uses a Content-Addressable Storage (CAS) model where all data is stored and retrieved using cryptographic hashes. This approach ensures data integrity and avoids duplication.

Core Components:

- **Object Store:**
Stores blobs, trees, and commits in a structured directory (.pes/objects) using hash-based naming and sharding.
- **Index (Staging Area):**
Maintains a record of files that are staged for the next commit.
- **Tree Logic:**
Represents directory structures and organizes files hierarchically.
- **Commit Logic:**
Creates snapshots of the project state and links them together to form a history.

Phase 1: Object Store Implementation

Goal:

To implement a content-addressable storage system with directory sharding.

Logic:

Each object is stored using a SHA-based hash. The first two characters of the hash form a directory, and the remaining characters form the filename. This prevents large directories and improves performance.

Screenshot 1A: Validation of object storage and retrieval.

```

aadya@aadya-VirtualBox:~/PES2UG24CS008-pes-vcs$ ./test_objects
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
Object stored at: .pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
Object stored at: .pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
PASS: deduplication
Object stored at: .pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
PASS: integrity check

All Phase 1 tests passed.

```

Screenshot 1B: Evidence of object files stored in a sharded directory structure.

```

aadya@aadya-VirtualBox:~/PES2UG24CS008-pes-vcs$ find .pes/objects -type f
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
aadya@aadya-VirtualBox:~/PES2UG24CS008-pes-vcs$ git add object.c

```

Phase 2: Tree Objects & Serialization

Goal:

To represent directory structures using tree objects.

Logic:

Tree objects store entries containing file mode, type, hash, and name. These objects are serialized into a compact binary format and stored in the object store.

Screenshot 2A: Output showing successful tree construction and serialization tests.

```

aadya@aadya-VirtualBox:~/PES2UG24CS008-pes-vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.

```

Screenshot 2B: Binary view of a tree object using xxd.

```

aadya@aadya-VirtualBox:~/PES2UG24CS008-pes-vcs$ find .pes/objects -type f
.pes/objects/2c/f8d83d9ee29543b34a87727421fdec7e3f3a183d337639025de576db9ebb4
aadya@aadya-VirtualBox:~/PES2UG24CS008-pes-vcs$ xxd .pes/objects/2c/f8d83d9ee29543b34a87727421fdec7e3f3a183d337639025de576db9ebb4 | head -20
00000000: 626c 6f62 2036 0068 656c 6c6f 0a      blob 6.hello.

```

Phase 3: Staging Area (Index)

Goal:

To implement a staging area that tracks files before committing.

Logic:

The index is maintained as a text file containing file metadata such as mode, hash, timestamp, and file path. It acts as an intermediate layer between the working directory and the repository.

Screenshot 3A: Execution of initialization, adding files, and checking status.

```
aadya@aadya-VirtualBox:~/PES2UG24CS008-pes-vcs$ rm -rf .pes
aadya@aadya-VirtualBox:~/PES2UG24CS008-pes-vcs$ ./pes init
Initialized empty PES repository in .pes/
aadya@aadya-VirtualBox:~/PES2UG24CS008-pes-vcs$ echo "hello" > file1.txt
aadya@aadya-VirtualBox:~/PES2UG24CS008-pes-vcs$ echo "world" > file2.txt
aadya@aadya-VirtualBox:~/PES2UG24CS008-pes-vcs$ ./pes add file1.txt file2.txt
Object stored at: .pes/objects/2c/f8d83d9ee29543b34a87727421fdec7e3f3a183d337639025de576db9ebb4
Object stored at: .pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f
aadya@aadya-VirtualBox:~/PES2UG24CS008-pes-vcs$ ./pes status
Staged changes:
  staged:    file1.txt
  staged:    file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked: test_objects
  untracked: commit.c
  untracked: test_tree.c
  untracked: pes.h
  untracked: object.c
  untracked: index.c
  untracked: test_tree
  untracked: Makefile
  untracked: index.h
  untracked: test_sequence.sh
  untracked: pes.c
  untracked: tree.c
  untracked: tree.h
  untracked: README.md
  untracked: commit.h
  untracked: test_objects.c
```

Screenshot 3B: Internal structure of the index file.

```
aadya@aadya-VirtualBox:~/PES2UG24CS008-pes-vcs$ cat .pes/index
100644 2cf8d83d9ee29543b34a87727421fdec7e3f3a183d337639025de576db9ebb4 1776654103 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776654114 6 file2.txt
aadya@aadya-VirtualBox:~/PES2UG24CS008-pes-vcs$
```

Phase 4: Commits, Logs, and References

Goal:

To create commits and maintain a history of changes.

Logic:

Each commit stores a reference to a tree object, a parent commit (if any), and metadata such as author and message. Commits are linked together to form a history chain.

Screenshot 4A: Output of commit history using pes log.

```
aadya@aadya-VirtualBox:~/PES2UG24CS008-pes-vcs$ ./pes init
Initialized empty PES repository in .pes/
aadya@aadya-VirtualBox:~/PES2UG24CS008-pes-vcs$ echo "Hello" > hello.txt
aadya@aadya-VirtualBox:~/PES2UG24CS008-pes-vcs$ ./pes add hello.txt
Object stored at: .pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
aadya@aadya-VirtualBox:~/PES2UG24CS008-pes-vcs$ ./pes commit -m "Initial commit"
Object stored at: .pes/objects/bd/9b769bb4d9910c31b4f0e99e4375fffea35c2055130c248ff131f78c626442
Object stored at: .pes/objects/75/3ac6c6e2dba2e102148a8f8396645fd57357cfd89ec0867a88ee783860c1c7
Committed: 753ac6c6e2db... Initial commit
aadya@aadya-VirtualBox:~/PES2UG24CS008-pes-vcs$ echo "World" >> hello.txt
aadya@aadya-VirtualBox:~/PES2UG24CS008-pes-vcs$ ./pes add hello.txt
Object stored at: .pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
aadya@aadya-VirtualBox:~/PES2UG24CS008-pes-vcs$ ./pes commit -m "Add world"
Object stored at: .pes/objects/d8/f28203071e10a3bde605928368bc5b51871287a867d8d10382967099dde814
Object stored at: .pes/objects/84/d048859fd0abba0ca42cb819d2b3a8cc944f0be56c599a7be017738071d8c7
Committed: 84d048859fd0... Add world
aadya@aadya-VirtualBox:~/PES2UG24CS008-pes-vcs$ echo "Goodbye" > bye.txt
aadya@aadya-VirtualBox:~/PES2UG24CS008-pes-vcs$ ./pes add bye.txt
Object stored at: .pes/objects/e6/7ed666cb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
aadya@aadya-VirtualBox:~/PES2UG24CS008-pes-vcs$ ./pes commit -m "Add farewell"
Object stored at: .pes/objects/a6/882ee4afdc855ea4e39f4e0fc77d90e73a6f72deb383e11934310a10a030f6
Object stored at: .pes/objects/5a/856376b993d466606f44e3d4f50b8abfc5c21ea9d1879c45bb21f53b13a651
Committed: 5a856376b993... Add farewell
aadya@aadya-VirtualBox:~/PES2UG24CS008-pes-vcs$ ./pes log
commit 5a856376b993d466606f44e3d4f50b8abfc5c21ea9d1879c45bb21f53b13a651
Author: Aadya Arvind <PES2UG24CS008>
Date: 1776654420

    Add farewell

commit 84d048859fd0abba0ca42cb819d2b3a8cc944f0be56c599a7be017738071d8c7
Author: Aadya Arvind <PES2UG24CS008>
Date: 1776654382

    Add world

commit 753ac6c6e2dba2e102148a8f8396645fd57357cfd89ec0867a88ee783860c1c7
Author: Aadya Arvind <PES2UG24CS008>
Date: 1776654352

    Initial commit
```

Screenshot 4B: Growth of objects in the .pes directory after multiple commits.

```
aadya@aadya-VirtualBox:~/PES2UG24CS008-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/5a/856376b993d466606f44e3d4f50b8abfc5c21ea9d1879c45bb21f53b13a651
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/75/3ac6c6e2dba2e102148a8f8396645fd57357cfd89ec0867a88ee783860c1c7
.pes/objects/84/d048859fd0abba0ca42cb819d2b3a8cc944f0be56c599a7be017738071d8c7
.pes/objects/a6/882ee4afdc855ea4e39f4e0fc77d90e73a6f72deb383e11934310a10a030f6
.pes/objects/bd/9b769bb4d9910c31b4f0e99e4375fffea35c2055130c248ff131f78c626442
.pes/objects/d8/f28203071e10a3bde605928368bc5b51871287a867d8d10382967099dde814
.pes/objects/e6/7ed666cb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/refs/heads/main
```

Screenshot 4C: Contents of HEAD and branch reference files.

```
aadya@aadya-VirtualBox:~/PES2UG24CS008-pes-vcs$ cat .pes/refs/heads/main
5a856376b993d466606f44e3d4f50b8abfc5c21ea9d1879c45bb21f53b13a651
aadya@aadya-VirtualBox:~/PES2UG24CS008-pes-vcs$ cat .pes/HEAD
ref: refs/heads/main
```

Final Integration

- **Full Integration Test:**

Confirms that all components of the system work together correctly.

```
aadya@aadya-VirtualBox:~/PES2UG24CS008-pes-vcs$ make test-integration
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Object stored at: .pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
Object stored at: .pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b9322919e9a23143d
Status after add:
Staged changes:
  staged:    file.txt
  staged:    hello.txt

Unstaged changes:
(nothing to show)

Untracked files:
(nothing to show)

--- First Commit ---
Object stored at: .pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafaa3829f61f75b8d10d5350f9
Object stored at: .pes/objects/16/48a1324e83be475dfaaee72620c40db60f5b746ad489b21536bb22878b4208
Committed: 1648a1324e83... Initial commit

Log after first commit:
commit 1648a1324e83be475dfaaee72620c40db60f5b746ad489b21536bb22878b4208
Author: Aadya Arvind <PES2UG24CS008>
Date: 1776654481

Initial commit
```

```
--- Second Commit ---
Object stored at: .pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff
Object stored at: .pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
Object stored at: .pes/objects/a8/b40a8537c0c8bcecb35a3eed26bc7ad34bb9843c197e2accb6ebd5a6cb0392
Committed: a8b40a8537c0... Update file.txt

--- Third Commit ---
Object stored at: .pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
Object stored at: .pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
Object stored at: .pes/objects/58/71056c3abc39e7ed517014ab16995bf66e070f855862fe70455f29382baa94
Committed: 5871056c3abc... Add farewell

--- Full History ---
commit 5871056c3abc39e7ed517014ab16995bf66e070f855862fe70455f29382baa94
Author: Aadya Arvind <PES2UG24CS008>
Date: 1776654481

Add farewell

commit a8b40a8537c0c8bcecb35a3eed26bc7ad34bb9843c197e2accb6ebd5a6cb0392
Author: Aadya Arvind <PES2UG24CS008>
Date: 1776654481

Update file.txt

commit 1648a1324e83be475dfaaee72620c40db60f5b746ad489b21536bb22878b4208
Author: Aadya Arvind <PES2UG24CS008>
Date: 1776654481

Initial commit

--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
5871056c3abc39e7ed517014ab16995bf66e070f855862fe70455f29382baa94
```

```
--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/16/48a1324e83be475dfaaee72620c40db60f5b746ad489b21536bb22878b4208
.pes/objects/58/71056c3abc39e7ed517014ab16995bf66e070f855862fe70455f29382baa94
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/a8/b40a8537c0c8bcecb35a3eed26bc7ad34bb9843c197e2accb6ebd5a6cb0392
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff

=== All integration tests completed ===
```

Technical Analysis

Section 5: Branching & Checkout Dynamics

Q5.1: How is pes checkout implemented and why is it complex?

The checkout operation changes the working directory to match a specific commit or branch. It involves updating the HEAD reference, reading the target commit, traversing its tree structure, and updating files accordingly. The complexity lies in ensuring that uncommitted changes are not lost and that file differences between branches are handled safely.

Q5.2: Detecting Dirty Working Directory Conflicts

Before switching branches, the system compares the current file state with the index and the target commit. If a file has been modified and also differs in the target branch, the operation is aborted to prevent overwriting user changes.

Q5.3: Detached HEAD State

A detached HEAD occurs when HEAD points directly to a commit instead of a branch. Commits can still be made in this state, but they are not associated with any branch. These commits can be recovered by creating a new branch pointing to them.

Section 6: Garbage Collection (GC) Strategy

Q6.1: Mark-and-Sweep Algorithm

Garbage collection can be implemented using a mark-and-sweep approach. First, all reachable objects are marked by traversing from references. Then, unmarked objects are deleted, ensuring only useful data remains.

Q6.2: Handling Race Conditions during GC

A race condition may occur if objects are created during garbage collection. To prevent accidental deletion, a grace period or timestamp check is used so that newly created objects are not removed prematurely.